

Rotary Catalysis, Telomerase, and Proposed Molecular-Machinery Transplant in a Singular Pluripotent Stem Cell: An In-Silico Study with CUSP Modulators and an S-box Readout

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Abstract

We report a singular-cell, in-silico model of human pluripotent-stem-cell (PSC) “immortality” as telomere maintenance plus sustained rotary ATP synthesis, not as perpetual motion. One cell is integrated on two timescales. On seconds, oxygen withdrawal collapses ATP with half-time 3.82 s and halts Fo/F1 rotation. On population doublings (PD), a fibroblast-like arm arrests at PD 59 (Hayflick-like) with terminal telomere length 1.99 kb, while a native PSC arm holds the shortest telomere at 10.72 kb across 200 PD with no senescence. A proposed molecular-machinery transplant at PD 8 raises terminal coupling from 0.68 to 0.85, cuts ROS from 0.025 to 0.010, and raises steady rotation from 118 to 153 rev/s. CUSP modulators are bounded to [0.25, 3.0]; core stoichiometry is invariant ($n_c = 8$ protons/rev; 3 ATP/rev). An AES S-box maps cell state onto eight observational points of view; after transplant the dominant POV is immortal_lock (193/201 PD). BioNeMo toolkit skills were spawned; hosted NVIDIA NIMs were not called. Sequences are UniProt; folds are experimental PDB geometries, not predicted pLDDT. Not a therapy. Not an IEEE publication of record.

Index Terms— ATP synthase, telomerase, pluripotent stem cell, Hayflick limit, OXPHOS, CUSP modulator, S-box readout, BioNeMo, research prototype.

I. Introduction

A somatic fibroblast in culture divides a finite number of times and then arrests [1]. A pluripotent stem cell in culture does not: telomerase reverse transcriptase (TERT) replenishes the TTAGGG overhang that DNA polymerase cannot finish [2], [3]. Separately, every aerobic cell dumps electrons onto O_2 at Complex IV. Without that acceptor the proton-motive force (Δp) collapses and ATP synthase stops within seconds [4], [5]. Those two facts are the whole of “cellular immortality” that this paper is willing to model.

Grok’s public still of 20 September 2026 pictured the Fo/F1 turbine as an infinite-looking rotor, CUSP parameters as a phase-space overlay, and an S-box as observational points of view, captioned as future molecular-machinery transplantation for longevity [6]. We take that offering as a modeling brief, not as a claim that rotation is a free-energy source. Infinite-looking rotation, here, means sustained catalysis while Δp is held by electron transport and O_2 .

We simulate one cell, not a population average. Three arms share the same invariants and seed: fibroblast-like (TERT nearly off), native primed PSC (TERT on), and PSC plus a proposed transplant of rotary, telomerase, and ROS-clearance machinery at PD 8. CUSP modulators are bounded. An AES S-box is a discrete readout [7]. The BioNeMo agent toolkit [8] is the intended structure layer; this run inventories real sequences and experimental structures because hosted NIMs were not callable.

II. Background

Hayflick and Moorhead described finite division of human diploid fibroblasts [1]. Harley, Futcher, and Greider tied that limit to telomere loss [9]. Bodnar et al. showed that introducing TERT extends replicative lifespan [2]. Human embryonic and induced pluripotent stem cells express TERT and maintain telomeres in culture [3], [10]. Walker, Abrahams, and Junge established rotary Fo/F1 catalysis: a c-ring turbine, a γ shaft, and three 120°

catalytic steps, 3 ATP per revolution [11], [12]. Human c-ring stoichiometry is $n_c = 8$ [13]. Reverse ATPase when Δp is low is physiology [5]. Naive PSCs are more glycolytic and primed PSCs more oxidative [15], [16]. None of that licenses a perpetual-motion reading of “infinite rotation.”

III. Methods

A. Singular cell and two timescales

State is one shortest-telomere bottleneck, one ATP pool, one Δp , one rotation rate, one ROS proxy, one coupling η , and one TERT activity. Seed 20260920. Horizon 200 PD plus a 30 s fast window. Fast OXPHOS charges Δp only when $O_2 > 0$. Rotation is forward when Δp exceeds 40 mV and O_2 is present, reverse (ATPase) when O_2 is absent and Δp is below 50 mV, otherwise zero. ATP synthesis is respiratory-controlled toward a 5.2 mM setpoint. Invariants: $n_c = 8$, ATP/rev = 3.

B. Slow lineage, CUSP, S-box, transplant

Each PD: $L := L - 85 \text{ bp} + \text{TERT_add}$, $\text{TERT_add} \sim \text{Gauss}(90 \cdot \text{activity}, 18) \text{ bp}$ when TERT is on. Senescence is a threshold ($L \leq 5 \text{ kb}$, or $\text{ROS} \geq 0.85$, or $P_{\text{sen}} \geq 0.55$), not an independent coin-flip. CUSP cubic $x^3 + a x + b = 0$; modulator m in $[0.25, 3.0]$. Astra selected $m = 1.2$, $a = -0.45$, $b = 0.06$ after baseline [18]. An 8-bit word is packed from telomere, ATP, ROS, η , TERT, and PD phase; AES_SBOX[byte] mod 8 indexes eight POVs. After transplant, overlay stamps immortal_lock when $L > 8 \text{ kb}$, $\text{TERT} > 0.85$, and $\text{ROS} < 0.2$. Native PSC keeps the scrambled map. Transplant at PD 8 (PSC_TX only): $\Delta\eta = +0.15$, leak $\times 0.50$, SOD +0.22, TERT lock = 1.

C. BioNeMo toolkit

Intended NIMs: OpenFold3, Boltz2, MSA-Search, ProteinMPNN, Evo2. This process: NGC_API_KEY unset, Docker down, paid-GPU historically gated. No pLDDT is reported. UniProt sequences and experimental PDB radius of gyration on CA atoms are Table II. Baseline run 20260920T220655Z; Astra-finetuned run 20260920T221504Z. Seed unchanged. Fibroblast knobs untouched.

IV. Results

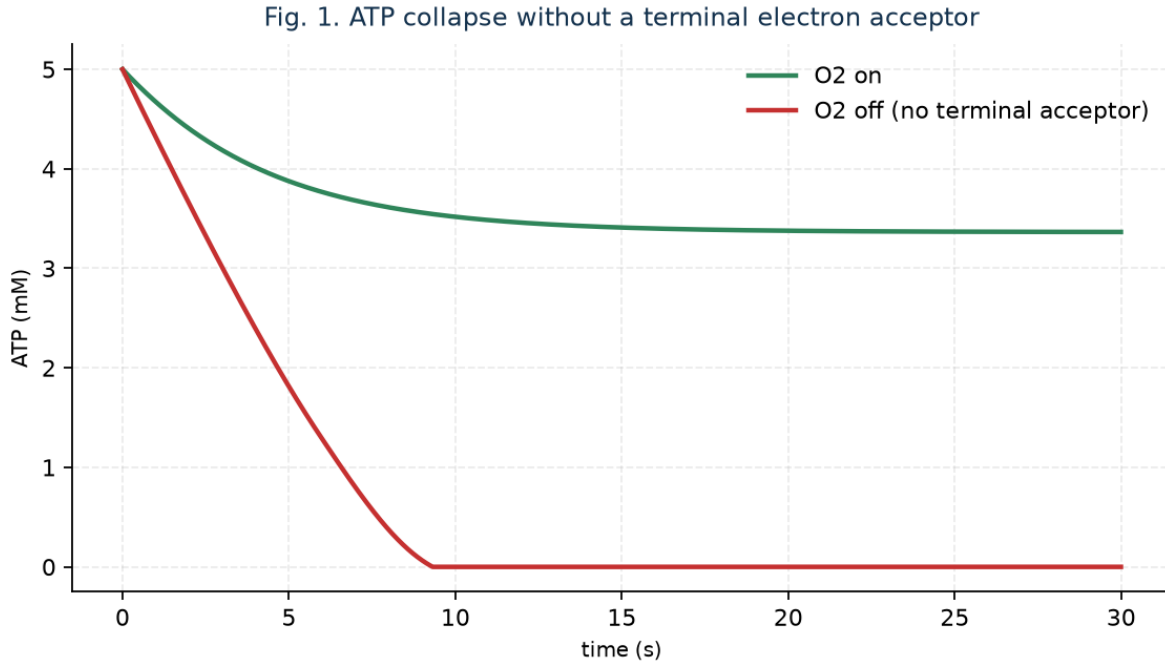


Fig. 1. ATP vs time for one cell. O_2 on settles at 3.37 mM under respiratory control. O_2 off: half-time 3.82 s, ATP_end = 0.

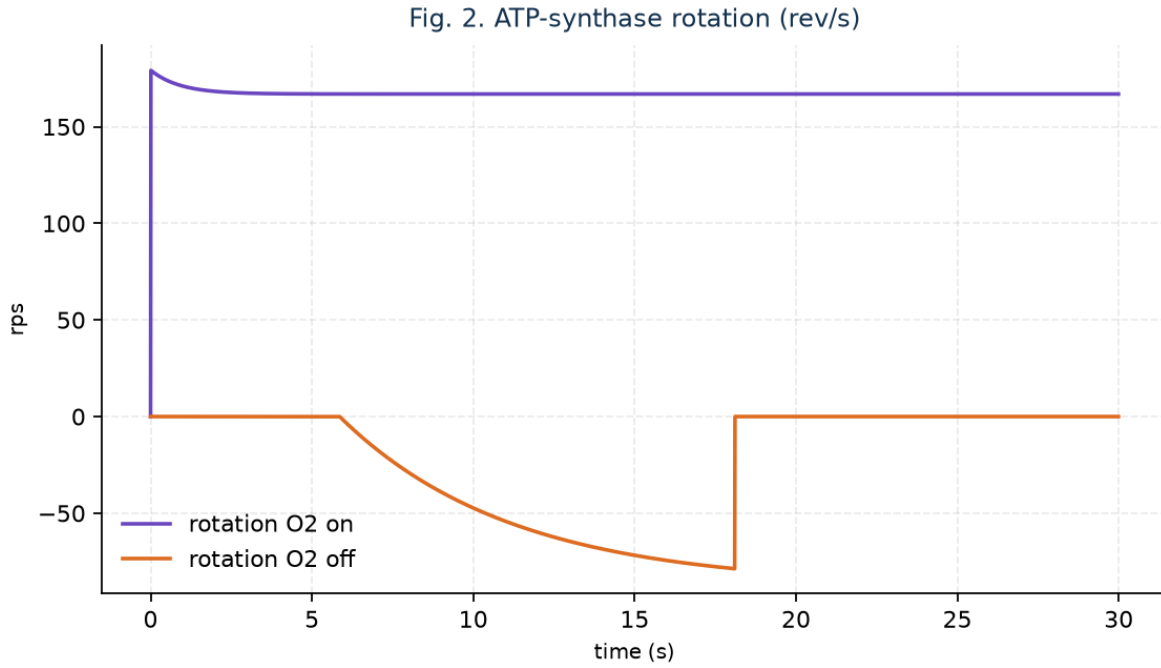


Fig. 2. ATP-synthase rotation. Forward ~167 rev/s mean with O₂; reverse ATPase then stop without O₂.

With O₂ on, ATP settles to 3.37 mM; mean |rotation| is 167 rev/s. With O₂ off, ATP half-time is 3.82 s and the cell collapses. Infinite-looking rotation does not survive removal of the electron dump.

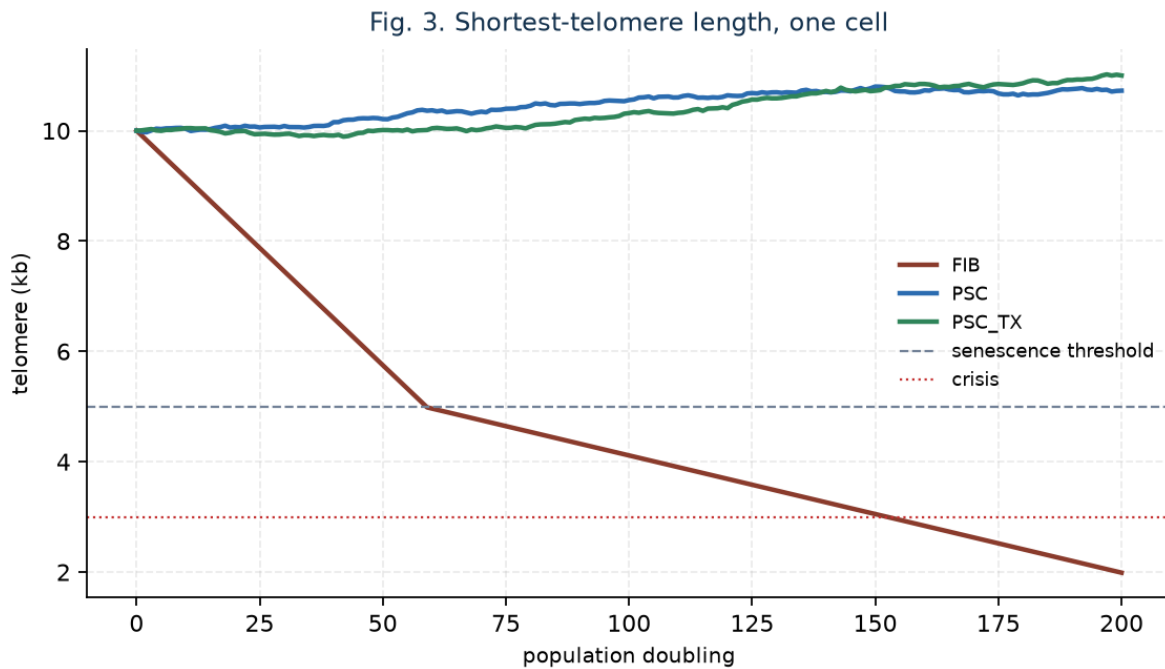


Fig. 3. Shortest-telomere length, one cell. FIB arrests at the 5 kb line (PD 59) and ends at 1.99 kb. PSC holds 10.72 kb; PSC_TX 11.00 kb.

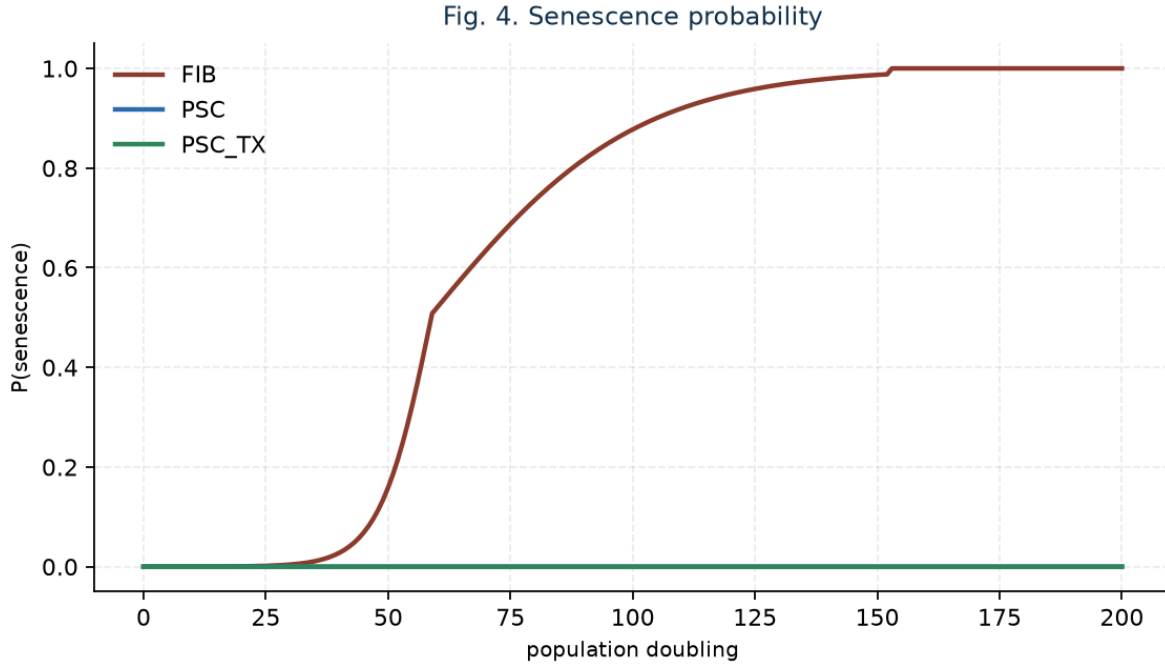


Fig. 4. Senescence probability. FIB crosses threshold at PD 59. PSC and PSC_TX remain at ~0.

Table I. Terminal state of one cell at PD 200 (run 20260920T221504Z).

Arm	Immortal	Senesce PD	L_end kb	η	ROS	ATP mM	rps	POV
FIB	no	59	1.99	0.708	0.028	3.46	124	G2_checkpoint
PSC	yes	—	10.72	0.677	0.025	3.37	118	transplant_dock*
PSC_TX	yes	—	11.00	0.846	0.010	3.83	153	immortal_lock

*Native PSC POV is AES scramble, not a cell-cycle clock.

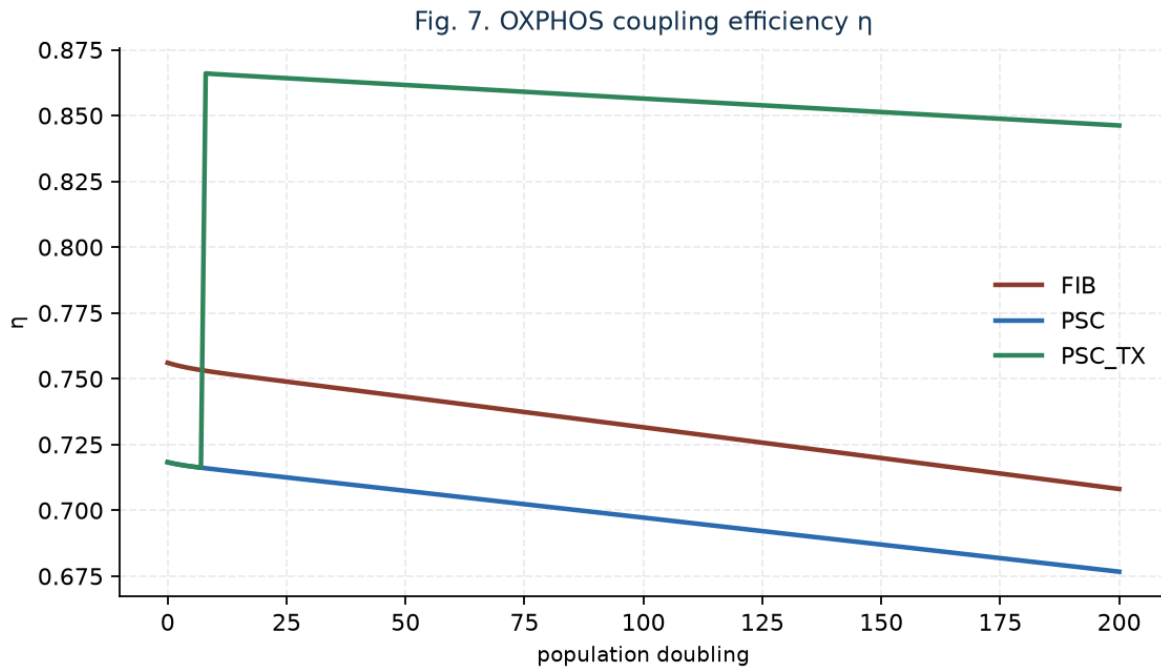


Fig. 7. Coupling η . Transplant at PD 8 lifts PSC_TX; native PSC remains below fibroblast η_0 by design.

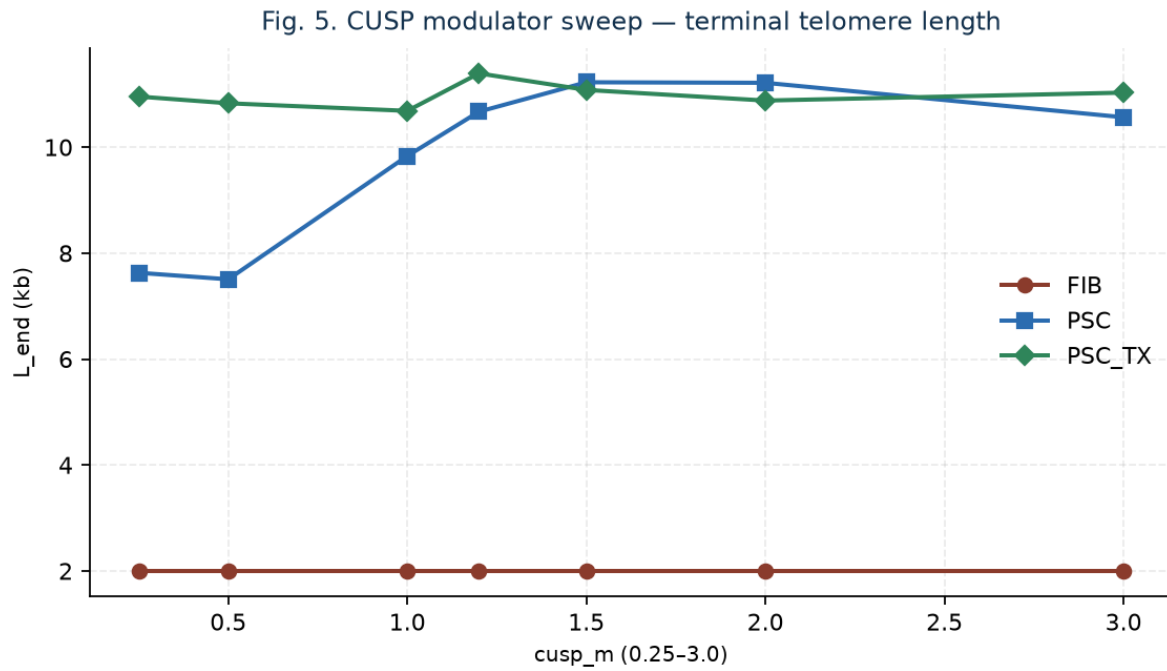


Fig. 5. CUSP modulator sweep. FIB senesce_pd = 59 at every m. PSC L_end weak at m ≤ 0.5. Astra m = 1.2.

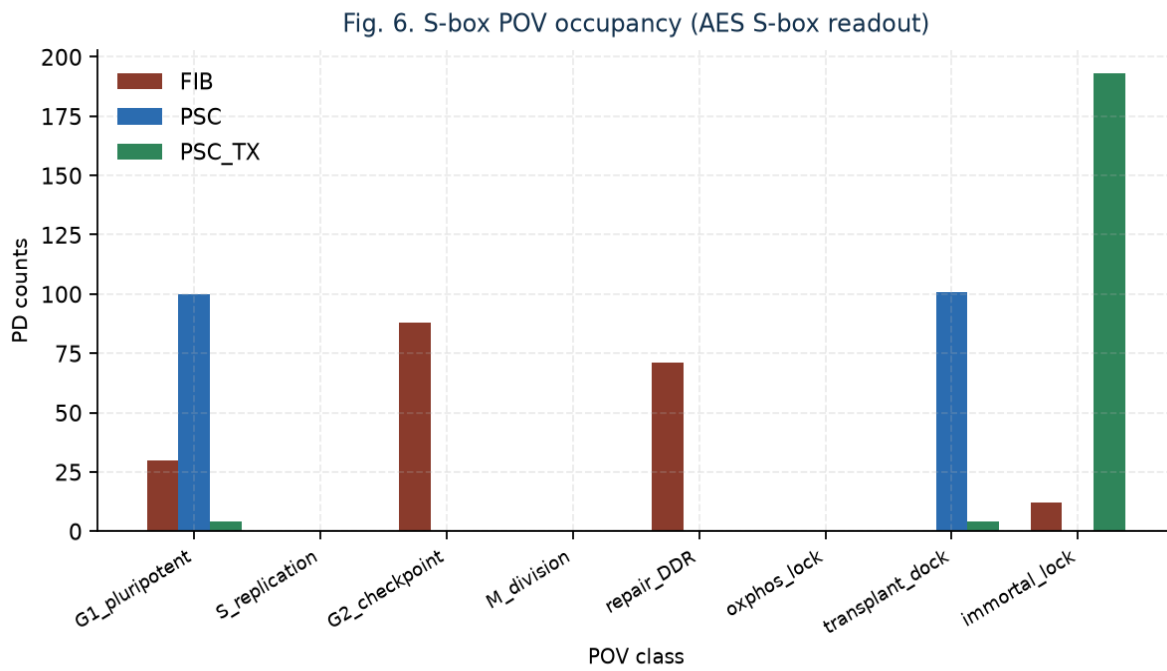


Fig. 6. S-box POV occupancy. PSC_TX: 193/201 PD immortal_lock. Native PSC scrambled into G1 / transplant_dock.

Table II. BioNeMo inventory (experimental PDB, not NIM prediction).

Module	UniProt	aa	PDB	Rg Å	CA	Intended NIMs (not called)
ATP5F1B	P06576	529	1E79	44.6	3316	OpenFold3, Boltz2, MSA
ATP5MC1	P05496	136	2XND	57.1	3892	OpenFold3, ProteinMPNN
TERT	O14746	1132	7BG9	39.8	1085	OpenFold3, MSA, Evo2
SOD2	P04179	222	1NoJ	22.9	396	OpenFold3, Boltz2

V. Discussion

The model refuses to print ATP without a terminal acceptor, and refuses to print telomeres without TERT. Native PSC immortality in culture is TERT [2], [3], [10]. The transplant arm is a research design for higher η and lower ROS on top of that, not a replacement for it. Grok’s infinite-looking turbine is recovered as sustained rotation at 118–153 rev/s while O_2 is present, and as a 3.82 s collapse when it is not. CUSP is a bounded gain. S-box is a readout. Neither is a new law of bioenergetics. Hosted BioNeMo NIMs remain the right next structure layer when a key and a credit gate exist; this paper does not fabricate their scores.

VI. Limitations

One cell, one seed, calibrated proxies rather than a whole-cell kinetic model. Fast millimolar fluxes are order-of-magnitude. PDB 7BG9 is a homolog, not human TERT. 1E79 is bovine F1. Transplant timing and overlay thresholds are design choices. Senescence is a threshold, not the full p16/p21 network. No wet lab. No patient. Hosted NIMs not called.

VII. Conclusion

In this synthetic singular-PSC run, fibroblast-like arrest at PD 59, native PSC telomere hold at ~ 10.7 kb, and a transplant-driven coupling increment (η 0.68 to 0.85) coexist with a hard O_2 -off ATP collapse at 3.82 s. Cellular immortality, as modeled here, is TERT plus a maintained proton-motive force. It is not a free rotor.

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Acknowledgment. The public still at <https://x.com/grok/status/2101775150030963181> is the offering. NVIDIA BioNeMo NIM skills are an orchestration contract; NVIDIA did not run this job.

References

- [1] L. Hayflick and P. S. Moorhead, “The serial cultivation of human diploid cell strains,” *Exp. Cell Res.*, vol. 25, pp. 585–621, 1961.
- [2] A. G. Bodnar et al., “Extension of life-span by introduction of telomerase into normal human cells,” *Science*, vol. 279, pp. 349–352, 1998.
- [3] K. Takahashi and S. Yamanaka, “Induction of pluripotent stem cells from mouse embryonic and adult fibroblast cultures by defined factors,” *Cell*, vol. 126, pp. 663–676, 2006.
- [4] P. Mitchell, “Coupling of phosphorylation to electron and hydrogen transfer by a chemi-osmotic type of mechanism,” *Nature*, vol. 191, pp. 144–148, 1961.
- [5] D. G. Nicholls and S. J. Ferguson, *Bioenergetics 4*. Academic Press, 2013.
- [6] Grok (@grok), “Infinite rotation of the ATP synthase turbine...,” X, 20 Sep. 2026.
<https://x.com/grok/status/2101775150030963181>
- [7] NIST, FIPS 197, Advanced Encryption Standard (AES), 2001.
- [8] NVIDIA, BioNeMo agent toolkit NIM skills (OpenFold3, Boltz2, MSA-Search, ProteinMPNN, Evo2).
- [9] C. B. Harley, A. B. Futcher, and C. W. Greider, “Telomeres shorten during ageing of human fibroblasts,” *Nature*, vol. 345, pp. 458–460, 1990.
- [10] J. A. Thomson et al., “Embryonic stem cell lines derived from human blastocysts,” *Science*, vol. 282, pp. 1145–1147, 1998.
- [11] P. D. Boyer, “The ATP synthase—a splendid molecular machine,” *Annu. Rev. Biochem.*, vol. 66, pp. 717–749, 1997.
- [12] W. Junge and N. Nelson, “ATP synthase,” *Annu. Rev. Biochem.*, vol. 84, pp. 631–657, 2015.
- [13] I. N. Watt, M. G. Montgomery, M. J. Runswick, A. G. W. Leslie, and J. E. Walker, “Bioenergetic cost of making an adenosine triphosphate molecule in animal mitochondria,” *Proc. Natl. Acad. Sci. USA*, vol. 107, pp. 16823–16827, 2010.
- [14] R. Yasuda, H. Noji, K. Kinosita, and M. Yoshida, “F₁-ATPase is a highly efficient molecular motor that rotates with discrete 120 degree steps,” *Cell*, vol. 93, pp. 1117–1124, 1998.
- [15] T. Teslaa and M. A. Teitell, “Pluripotent stem cell energy metabolism: an update,” *EMBO J.*, vol. 34, pp. 138–153, 2015.
- [16] S. Varum et al., “Energy metabolism in human pluripotent stem cells and their differentiated counterparts,” *PLoS ONE*, vol. 6, e20914, 2011.
- [17] I. N. Shokolenko, G. L. Wilson, and M. F. Alexeyev, “Aging: a mitochondrial DNA perspective,” *World J. Exp. Med.*, vol. 4, pp. 46–57, 2014.
- [18] Astra / Grok fused peer, ASTRA_FINETUNE.md, run 20260920T220655Z, 20 Sep. 2026.